

Q1:

10010101

$$R_1 = 0.08$$

$$R_2 = 0.79$$

$$0.08 \times 8 = 0.64 \approx 1$$

$$0.79 \times 8 = 6.32 \approx 6$$

10010101

10010101

10010101

3 chromosomes select correct 3
one select 2.5
E.C. = 1

Premature -

- 1) All solution similar
- 2) Always select best one
- 3) Low mutation rate
solution

- 1) Increase Mutation rate
- 2) Not Always select best / we have rank
- 3) Less Elitism
- 4) Restart partially
- 5) Random Individual Add Pop

C1	4	0.1818	0 — 0.1818
C2	7	0.318	0.19 — 0.4998
C3	2	0.090	0.4999 — 0.5898
C4	9	0.4090	0.5899 — 1
	<u>22</u>		

0.82 → select C4

0.69 — select C4 duplicate Not Allow

0.57 — select C3

C3 & C4

C2	10	10	10	10	$R_1 = 0.24$
C4	01	10	01	11	$R_2 = 0.62$

$0.24 \times 8 = 1.92 \approx 2$

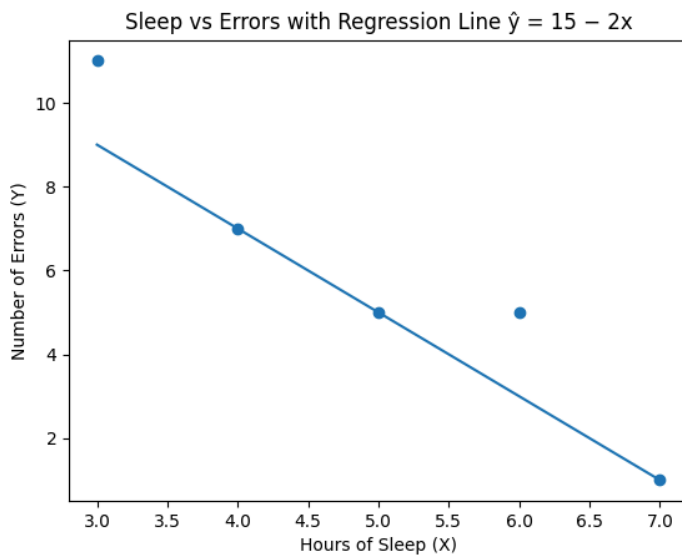
$0.62 \times 8 = 4.96 \approx 5$

osp1 10 100 010

osp2 01 101 ~~111~~

Q2:

(a)



(b)

Residuals are small (0 or +2) suggesting a good fit. Positive residuals suggest slight

Participant	X	Actual Y	Predicted $\hat{y} = 15 - 2x$	Residuals/Errors ($Y - \hat{Y}$)
P1	7	1	1	0
P2	3	11	9	+2
P3	6	5	3	+2
P4	4	7	7	0
P5	5	5	5	0

underestimation for some points.

(c)

$R^2 = 0.82$ means 82% of the variation in the errors on the cognitive test is explained by the sleep hours by the given regression model. It indicates a strong model fit.

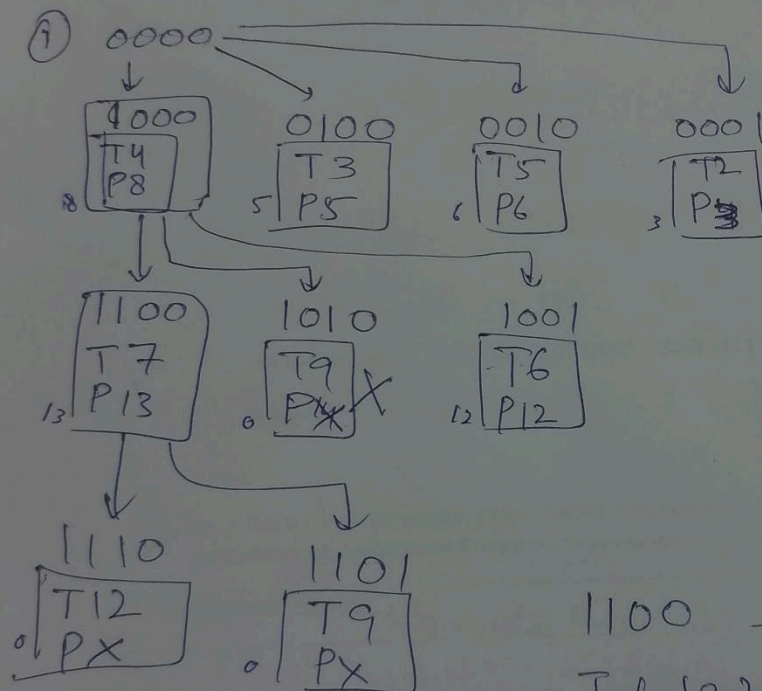
$r = -0.91$ shows a strong negative correlation. As sleep hours increase, the errors on the test decrease significantly.

Q3:

National University of Computer and Emerging Sciences
Lahore Campus

Task Number	1	2	3	4
Profit	8	5	6	3
Time	4	3	5	2

For the set of four tasks shown above and a system with maximum **time capacity 8**, start from the initial solution **0000**, where each bit represents whether a task is selected or not selected. Apply the hill-climbing (local search) algorithm to find the best solution. Assume that the successors/neighbours of a solution are created by setting exactly one of the zero bits to 1. You must show all intermediate solutions generated in each iteration. The process should stop when no neighbouring solution yields a better fitness value than the current solution.



1100 final solution
Task 122
Profit 13
Time 7